Copper

Author: Usense Healthcare

Identity Sheet

Property	Value
Name	Copper
Other names	Cu, Cu2+, cupric ion
Chemical formula	Cu^{2+} (predominant biological form)
Atomic weight	63.55 g/mol
CAS number	7440-50-8
PubChem CID	23978
Appearance	Colourless to blue-green in solution (Cu2+ complexes)

Structural representation:

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Cu(2+) (transition metal divalent cation)
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Copper is an essential trace element, serving as a cofactor for numerous redox enzymes (cytochrome c oxidase, superoxide dismutase, ceruloplasmin, lysyl oxidase, dopamine beta-hydroxylase). Total body copper: ~75-150 mg. Copper homeostasis is tightly regulated via intestinal absorption (ATP7A) and hepatic biliary excretion (ATP7B). Urinary copper is normally very low; elevated urinary copper is the hallmark of **Wilson disease**.

Molecules Not to Be Confused With

Molecule	Formula	MW	Key Difference
Ceruloplasmin	Protein	~132 kDa	Copper-binding plasma protein; carries ~90% of serum Cu; low in Wilson disease
Zinc (Zn2+)	Zn^{2+}	65.38	Another essential trace metal; competes with Cu absorption
Iron (Fe2+/Fe3+)	Fe	55.85	Different transition metal; different metabolic roles

Medical Information

Origin

Endogenous Copper is not synthesised. Total body content: ~75-150 mg. Liver is the main storage organ. Ceruloplasmin (synthesised in liver) carries ~90% of plasma copper.

Exogenous Dietary sources: shellfish, liver, nuts, seeds, whole grains, dark chocolate, mushrooms. RDA: 0.9 mg/day (adults). Typical intake: 1-2 mg/day. Absorbed in duodenum/jejunum via CTR1 transporter; regulated by ATP7A (Menkes protein).

Primary & Secondary Biological Roles

Primary role: - **Enzymatic cofactor:** Essential for cytochrome c oxidase (ETC Complex IV), Cu/Zn-SOD (antioxidant), ceruloplasmin (ferroxidase, iron metabolism), lysyl oxidase (collagen/elastin crosslinking), dopamine beta-hydroxylase (catecholamine synthesis), tyrosinase (melanin).

Secondary roles: - **Iron metabolism:** Ceruloplasmin oxidises Fe2+ to Fe3+ for transferrin loading. - **Connective tissue integrity:** Lysyl oxidase requires Cu for collagen/elastin crosslinking. - **Immune function:** Cu deficiency impairs neutrophil function.

Catabolism and Elimination Pathway

- **Biliary excretion** is the primary route (~80%); regulated by ATP7B in hepatocytes.
- Urinary copper excretion is normally very low (<40 ug/day) – Cu is protein-bound (ceruloplasmin, albumin) and not freely filtered.
- In Wilson disease, ATP7B dysfunction leads to copper accumulation in liver and overflow into blood (free Cu) and urine.
- D-penicillamine chelates Cu and dramatically increases urinary Cu excretion (used diagnostically and therapeutically).

Expression in Humans

Normal Levels

Compartment	Reference Range
Serum copper	70-140 ug/dL (11-22 umol/L)
Serum ceruloplasmin	20-50 mg/dL
Urinary copper (24-h)	<40 ug/day (<0.6 umol/day)
Urinary Cu (Wilson disease)	>100 ug/day (often >200 ug/day)
Post-penicillamine urinary Cu	>1600 ug/day diagnostic for Wilson
Hepatic copper	<50 ug/g dry weight (normal); >250 ug/g (Wilson)

Factors Influencing Levels **Increased urinary copper:** - Wilson disease (ATP7B mutations) - D-penicillamine or trientine chelation therapy - Acute hepatitis (hepatocyte necrosis releases Cu) - Nephrotic syndrome (ceruloplasmin loss; free Cu filtered) - Copper poisoning (rare; occupational) - Chronic active hepatitis

Decreased urinary copper: - Normal state - Menkes disease (ATP7A mutation; copper malabsorption; very low serum and urine Cu)

Associated Pathologies

Condition	Copper Pattern	Key Symptoms
Wilson disease	Low ceruloplasmin, high urine Cu (>100 ug/day), high hepatic Cu	Liver disease, neuropsychiatric symptoms, Kayser-Fleischer rings; 1:30,000
Menkes disease	Very low serum Cu, low ceruloplasmin, low urine Cu	Kinky hair, seizures, connective tissue defects; X-linked; ~1:100,000

Condition	Copper Pattern	Key Symptoms
Copper deficiency	Low serum Cu	Anaemia, neutropenia, myelopathy
Copper toxicity	Very high serum Cu, high urine Cu	Nausea, liver failure (acute); rare

Presence in Urine

Should it be normally present? In **trace amounts** only (<40 ug/day). Most copper is protein-bound and not filtered.

Normal urinary levels: <40 ug/day (<0.6 umol/day).

Form in urine: Cu^{2+} ion, partially complexed with amino acids and small organic molecules.

Pathological significance:

Urinary Cu	Possible Causes	Prevalence
>100 ug/day	Wilson disease, chelation therapy	Wilson: ~1:30,000
<10 ug/day	Normal or Menkes disease	–

Solubility: Cu2+ salts generally soluble. No urinary crystal issues.

Detection in Urine

Available Clinical Assays

1. **Flame or graphite furnace AAS:**
   - **Principle:** Cu atoms absorb light at **324.7 nm** from a Cu hollow-cathode lamp.
   - **Detection:** Absorption at **324.7 nm**.
   - **LOD:** ~0.5 ug/L (GFAAS); ~5 ug/L (flame AAS).
   - **Advantages:** Reference method; highly specific.
2. **ICP-OES / ICP-MS:**
   - **Principle:** Plasma excitation/ionisation of Cu.
   - **Detection:** Emission at 324.7/327.4 nm (OES) or m/z 63/65 (MS).
   - **LOD:** ~0.01 ug/L (ICP-MS).
   - **Advantages:** Multi-element; ultimate sensitivity.
3. **Colorimetric (bathocuproine/BCS):**
   - **Principle:** Cu+ (after reduction) reacts with bathocuproine disulfonate -> orange complex.
   - **Detection:** Absorbance at **484 nm**.
   - **LOD:** ~1 ug/dL.
   - **Advantages:** Simple, automatable.

Optimal Urine Type for Measurement

- **24-hour urine** in acid-washed, metal-free containers (trace element contamination is the main preanalytical concern).
- Acidify with 6N HCl (prevents Cu adsorption to container walls).
- Avoid rubber stoppers, coloured containers.
- Random specimens unreliable due to variable concentration.

Actual Gold Standard

ICP-MS is the reference method for trace copper in urine. GFAAS is widely used in clinical labs. CV: 5-10%.

Interferences in Measurement

Interference	Effect	Mechanism
Container contamination	Falsely high	Metal from container, stoppers
Haemolysis	Positive bias	Erythrocyte Cu release
EDTA anticoagulant	Negative bias if chelated	Chelates Cu
High zinc	Spectral interference (AAS)	Adjacent lines (AAS); resolved by ICP-MS

Research Detection Methods

Spectroscopy Detection (UV-Vis / NIR)

- **AAS at 324.7 nm:** See clinical assay. Reference.
- **UV-Vis colorimetric (bathocuproine, 484 nm):** LOD ~1 ug/dL.
- **Cu2+ d-d transitions:** Very weak absorption at ~600-900 nm (blue-green); not useful for trace analysis.

Fluorescence Detection

- Cu2+ is a fluorescence quencher (paramagnetic d9). Indirect approaches:
- **Turn-off probes:** Fluorescent chelators quenched by Cu2+ (e.g., calcein: Ex 490/Em 515 nm). LOD: ~0.1 umol/L.
- **Turn-on probes:** Cu2+-triggered dequenching or reaction-based probes. Various Ex/Em. LOD: ~0.01-0.1 umol/L.
- **Quantum dots (CdTe):** Cu2+ quenches QD fluorescence. Ex 350/Em 540 nm. LOD: ~5 nM.

Raman Detection

- Cu2+ (monatomic ion) has no Raman modes. Not directly detectable.
- **Cu complexes** with organic ligands have Raman-active modes.

FTIR Detection

- Not applicable for Cu2+ ion detection.

Voltammetry Detection

- **Anodic stripping voltammetry (ASV):** Cu is deposited on electrode at -0.4 V, then stripped (oxidised) at ~+0.05 V vs Ag/AgCl. Peak current proportional to Cu concentration.
  - LOD: ~0.1-1 ug/L (ppb level).
  - **Electrode:** Mercury film (HMDE), bismuth film, or carbon.
  - **Advantages:** Very sensitive, inexpensive, field-deployable.
- **Potentiometric Cu-ISE:** LOD ~0.01 umol/L.

Other Detection Technologies

1. **Total reflection XRF (TXRF):** LOD ~0.1 ug/L. Multi-element, no digestion needed.
2. **Neutron activation analysis (NAA):** LOD ~0.01 ug/L. Ultimate reference. Limited access.
3. **Paper-based colorimetric:** Dithizone or bathocuproine on paper. LOD: ~50 ug/L.

Summary Table: Detection Methods Comparison

Method	LOD	Key Parameter	Sample Prep	Strengths
GFAAS	~0.5 ug/L	324.7 nm absorption	Acid-washed	Reference
ICP-MS	~0.01 ug/L	m/z 63/65	Dilution	Multi-element, ultimate
Colorimetric (BCS)	~10 ug/L	484 nm	Reduction	Simple
ASV	~0.1 ug/L	Strip at +0.05 V	Deposition	Sensitive, cheap
QD fluorescence	~5 nM	Ex 350/Em 540 nm	Probe	Sensitive
Cu-ISE	~0.01 umol/L	Potentiometric	None	Real-time

References

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